



Object-Oriented Programming review lab

Title: HTU Smart Library System

At **Al Hussein Technical University (HTU)**, the library was once a quiet place where students could focus on their studies. Lately, however, the library had become overcrowded and chaotic. Students would often wander between shelves, searching for books or digital materials, only to find that many items were either missing or misplaced.

Ms. Rania, the head librarian, saw the growing frustration among students and realized it was time to modernize the library. She wanted a **smart system** that could keep track of all items, help students find what they needed quickly, and make borrowing simpler. But instead of purchasing expensive software, she had an idea: why not involve HTU's **Computer Science students**? This would be a perfect opportunity for them to practice Object-Oriented Programming (OOP) in a real-world scenario.

The challenge was clear: **create a console application in C#** that simulates a "Smart Library System" using all major OOP principles.

The System

The system would manage two types of library items:

1. **Books**: Physical books with pages, and categories.
2. **E-Books**: Digital books that could be downloaded or read online, including file size and download links.

At the heart of the system was an **abstract class** called `LibraryItem`, representing any item in the library. Every item had a **title, author, year of publication, and availability status**. The system enforced **encapsulation**, making all attributes private and accessible only through validated getters and setters.

Each item type had its own **specialized class** derived from `LibraryItem`:

- **Book**: Added `NumberOfPages` and `Category`. Overridden `DisplayInfo()` showed detailed book information.
- **EBook**: Added `FileSizeMB` and `DownloadLink`. Overridden `DisplayInfo()` displayed digital book details.

The system also supported **searching** using the overloaded Matches() method—by title alone or by both title and author that should be exist in the LibraryItem class.

The Librarian

In this system, the **Library Manager** acted as the librarian. Just like Ms. Rania in real life, the manager could:

- Add new books, or e-books.
- Search for items using the overloaded search methods.
- Display all items in the library, using **polymorphism** to call the correct DisplayInfo() method for each object.

The Main Program

A welcoming menu appeared:

Welcome to HTU Smart Library System

1. Add new Book
2. Add new E-Book
3. Show all items
4. Search by title
5. Search by title and author
6. Exit

Console Application example:

```
=== HTU Smart Library System ===
1. Add new Book
2. Add new E-Book
3. Show all items
4. Search by title
5. Search by title and author
6. Exit
Choose an option: 1
Enter Book Title: Guide to Competitive programming
Enter Author: Antti Laaksonen
Enter Year: 2024
Is this book available? write Yes/yes or Noyes
Enter Number of Pages: 338
Enter Category: Science and technology

=== HTU Smart Library System ===
1. Add new Book
2. Add new E-Book
3. Show all items
4. Search by title
5. Search by title and author
6. Exit
Choose an option: 2
Enter E-Book Title: CP Handbook
Enter Author: Felix Halim
Enter Year: 2013
Enter File Size (MB): 3564
Enter Download Link: https://handbook.com
Is this EBook available? write Yes/yes or Nono
```

```
=== HTU Smart Library System ===
1. Add new Book
2. Add new E-Book
3. Show all items
4. Search by title
5. Search by title and author
6. Exit
Choose an option: 3
Book information:
Author Antti Laaksonen
Title Guide to Competitive programming
Number of pages 338
Publish year 2024
Book category Science and technology

EBook information:
Author Felix Halim
Title CP Handbook
File size 3564
Publish year 2013
Download link https://handbook.com
```

Show the library item details if it exists

```
=== HTU Smart Library System ===
1. Add new Book
2. Add new E-Book
3. Show all items
4. Search by title
5. Search by title and author
6. Exit
Choose an option: 4
Enter title to search: CP Handbook
EBook information:
Author Felix Halim
Title CP Handbook
File size 3564
Publish year 2013
Download link https://handbook.com
```